Academic Projects

Jailbreak Prompt Analysis in Large Language Models – Information Assurance (Jan 2025 – May 2025)

- Conducted adversarial testing on **LLMs (GPT-4, PaLM-2)** to analyze jailbreak prompt strategies, achieving over **90% success rate** in bypassing safety filters using narrative misdirection and hybrid prompt engineering.
- Developed an automated prompt mutation framework using Python and OpenAI API, **converting 729 out of 766 failed prompts into successful jailbreaks**.
- Evaluated model robustness through custom metrics like Jailbreak Success Rate (JSR) and Expected Maximum Harmfulness (EMH), supporting cybersecurity-focused AI safety research.

Smart Hire Solutions (AI-Driven Job Matching Platform) – Advanced Software Engineering – (Aug 2024 – Dec 2024)

- Led a 10-member team to develop a scalable job-matching platform utilizing Node.js, Express.js, and MongoDB, effectively managing over 10,000 user profiles and resumes.
- Engineered RESTful APIs for user authentication, profile management, and resume parsing, enhancing system efficiency and reducing manual screening time by up to 60%.
- Integrated automated job scraping and real-time data ingestion, delivering personalized job recommendations and maintaining 91.9% system uptime.

Fibonacci Heap Implementation – Design and Analysis of Algorithms (Aug 2024 – Dec 2024)

- Developed a complete Fibonacci Heap from scratch using Python and Java, demonstrating amortized O (1) time complexity for insertion and merge, and O (log n) for extract-min operations.
- Benchmarked performance against Binary and Binomial Heaps across 100,000+ operations, achieving up to 40% faster decrease-key efficiency in Dijkstra and Prim’s algorithms.
- Validated correctness with unit tests (JUnit) and applied the heap structure to optimize graph-based shortest path and spanning tree problems.

Distributed Messaging System – Advanced Operating Systems (Jan 2024 – May 2024)

- Engineered a **highly scalable, reliable, and fault-tolerant** messaging system capable of handling **millions of messages per day**, using technologies like Apache Kafka, RabbitMQ, and Apache Pulsar.
- Designed the system to efficiently distribute tasks across **multiple nodes**, improving system resiliency and uptime for real-world applications in **business messaging**, **real-time communication**, and **IoT platforms**.
- Optimized message processing speed, reducing latency by **25%** under heavy load conditions

Cross Entropy Based English Phrase Identification System – Human Computer Interaction (Jan 2024 – May 2024)

- Enhanced a machine learning model for part-of-speech tagging, achieving a **15% improvement** in accuracy for classifying English words by grammatical correctness.
- Revamped the model's detection capabilities by implementing advanced techniques such as maximum statistics and smoothing, which resulted in a **40% increase in accuracy** for identifying valid ('V') and non-detection ('N') phrases.
- Improved classification of unknown features by **20%** through innovative smoothing techniques, increasing the model's robustness on unseen datasets.

Driver Drowsiness Detection - Computer Vision (Jan 2024 – May 2024)

- Developed a real-time drowsiness detection system using OpenCV and Python, achieving **92% accuracy** in detecting prolonged eye closure from video streams.
- Implemented Haar cascade classifiers and eye aspect ratio tracking across consecutive frames to identify fatigue symptoms and trigger timely alerts.
- Processed live webcam input to analyze driver eye behaviour, reducing false positives by **30%** through temporal smoothing and blink duration analysis.

House price prediction - Machine Learning (Aug 2023 – Dec 2023)

- Developed a predictive model using Python and linear regression, achieving **90% accuracy** in forecasting house prices based on historical and real-time data.
- Applied the model to enable real estate agents, homeowners, and buyers to make informed decisions by accurately predicting property values, reducing price estimation errors by **10%** compared to traditional methods.
- Demonstrated data science techniques with a focus on real estate applications, providing actionable insights for the industry.

Sorting Algorithm Performance Analysis – Design and Analysis of Algorithms (Jan 2023 – May 2023)

- Implemented and evaluated 7 sorting algorithms (Merge Sort, Heap Sort, Quick Sort, etc.) in Java across datasets up to **10,000 elements**, measuring execution time and efficiency.
- Generated comparative performance graphs to identify optimal algorithms under varying input conditions, achieving up to **60% improvement** in time efficiency using median-of-three Quick Sort.
- Conducted empirical complexity analysis and visualized algorithm behaviour using runtime data, aiding in practical understanding of time-space trade-offs.

ABC Inventory Classification Using Machine Learning – Data Analysis & Modelling Techniques (Jan 2023 – May 2023)

- Developed a machine learning system to automate multi-criteria ABC inventory classification using algorithms like SVM, k-NN, and Backpropagation Neural Networks.
- Achieved **over 92% classification accuracy** with SVM, outperforming traditional MDA methods by up to **18%** across benchmark datasets.
- Integrated statistical analysis and CSV-based datasets to enhance inventory decision-making in ERP systems, demonstrating scalable ML-based inventory optimization.